

Univessel Glass 10 L - 10 L vessel

sartorius-univessel-glass-10

Field-level manufacturer evidence; remaining values unverified

Each field is classified independently. Direct manufacturer values take precedence over estimates only for the matching model and configuration. Approximate values printed by the manufacturer remain approximate. Unverified fields are editable simulation assumptions, not equipment operating limits.

Parameter	Value in simulator	Evidence status
Minimum working volume	1.5 L	Source-supported
Maximum working volume	10 L	Source-supported
Nominal vessel / bag size	10 L	Source-supported
Total vessel volume loaded into the form	13 L	Source-supported
Initial working volume loaded into the form	1.5 L	Derived / estimated
Product-contact material	Borosilicate glass / AISI 316L stainless steel / EPDM	Source-supported
Height : diameter	1.9	Source-supported
Internal / chamber diameter	0.19 m	Source-supported
Liquid height at maximum working volume	0.36 m	Source-supported
Impeller / mixing type	Pitched-blade turbine	App assumption
Impeller count	2	App assumption
Impeller diameter	0.078 m	Source-supported
Power number	1.3	App assumption
Initial agitation	350 rpm	App assumption
Maximum agitation	1200 rpm	App assumption
Initial aeration	0.3 vvm	App assumption
Maximum aeration	2 vvm	App assumption
Baffled vessel	Yes	App assumption
Maximum pressure	0.5 bar(g)	App assumption
Default operating pressure	0 bar(g)	App assumption
Maximum inlet oxygen fraction	100%	App assumption

Parameter evidence and derivations

Minimum working volume - 1.5 L

Source-supported | Sartorius Univessel Glass brochure SBI1554-e - PDF page 8; Inside Dimensions, 10 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 1.5 L.

Limitations: Selected 3-blade segment-impeller configuration. Published liquid H/D is rounded; direct dimensions take precedence over recomputation. Millimetres converted to metres.

Local original: [Sartorius Univessel Glass brochure SBI1554-e - PDF page 8](#)

[Manufacturer publication](#)

Maximum working volume - 10 L

Source-supported | Sartorius Univessel Glass brochure SBI1554-e - PDF page 8; Inside Dimensions, 10 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 10 L.

Limitations: Selected 3-blade segment-impeller configuration. Published liquid H/D is rounded; direct dimensions take precedence over recomputation. Millimetres converted to metres.

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Nominal vessel / bag size - 10 L

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Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 10 L.

Limitations: Selected 3-blade segment-impeller configuration. Published liquid H/D is rounded; direct dimensions take precedence over recomputation. Millimetres converted to metres.

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Total vessel volume loaded into the form - 13 L

Source-supported | Sartorius Univessel Glass brochure SBI1554-e - PDF page 8; Inside Dimensions, 10 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 13 L.

Limitations: Selected 3-blade segment-impeller configuration. Published liquid H/D is rounded; direct dimensions take precedence over recomputation. Millimetres converted to metres.

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Initial working volume loaded into the form - 1.5 L

Derived / estimated | Form-population rule

Rule: initial working volume = min(minimum working volume, maximum working volume)

Selection / calculation: $\min(1.5 \text{ L}, 10 \text{ L}) = 1.5 \text{ L}$

Limitations: The preset starts the process at the lower end of its configured working range.

Product-contact material - Borosilicate glass / AISI 316L stainless steel / EPDM

Source-supported | Sartorius Univessel Glass brochure SBI1554-e - PDF page 6; Materials of major product wetted parts

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: Borosilicate glass / AISI 316L stainless steel / EPDM.

Limitations: Major wetted parts, not just the glass vessel shell.

Local original: [Sartorius Univessel Glass brochure SBI1554-e - PDF page 6](#)

[Manufacturer publication](#)

Height : diameter - 1.9

Source-supported | Sartorius Univessel Glass brochure SBI1554-e - PDF page 8; Inside Dimensions, 10 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 1.9.

Limitations: Selected 3-blade segment-impeller configuration. Published liquid H/D is rounded; direct dimensions take precedence over recomputation. Millimetres converted to metres.

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Internal / chamber diameter - 0.19 m

Source-supported | Sartorius Univessel Glass brochure SBI1554-e - PDF page 8; Inside Dimensions, 10 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.19 m.

Limitations: Selected 3-blade segment-impeller configuration. Published liquid H/D is rounded; direct dimensions take precedence over recomputation. Millimetres converted to metres.

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Liquid height at maximum working volume - 0.36 m

Source-supported | Sartorius Univessel Glass brochure SBI1554-e - PDF page 8; Inside Dimensions, 10 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.36 m.

Limitations: Selected 3-blade segment-impeller configuration. Published liquid H/D is rounded; direct dimensions take precedence over recomputation. Millimetres converted to metres.

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Impeller / mixing type - Pitched-blade turbine

App assumption | Sartorius Univessel Glass brochure SBI1554-e - PDF page 8; Inside Dimensions, segment impeller

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: Pitched-blade turbine.

Limitations: Actual hardware: 3-blade segment impeller. Pitched-blade is the simulator approximation; its generic power number is not a manufacturer measurement.

Local original: Sartorius Univessel Glass brochure SBI1554-e - PDF page 8

Manufacturer publication

Impeller count - 2

App assumption | Generic reusable glass profile

Rule: `PROFILE_DEFAULTS["glass"].impellerCount = 2`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable glass starting profile. The count is used by the vessel diagram and represents a generic configuration unless source-supported.

Impeller diameter - 0.078 m

Source-supported | Sartorius Univessel Glass brochure SBI1554-e - PDF page 8; Inside Dimensions, 10 L column

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 0.078 m.

Limitations: Selected 3-blade segment-impeller configuration. Published liquid H/D is rounded; direct dimensions take precedence over recomputation. Millimetres converted to metres.

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Manufacturer publication

Power number - 1.3

App assumption | Generic reusable glass profile

Rule: PROFILE_DEFAULTS["glass"].powerNumber = 1.3

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable glass starting profile. The power number is an empirical generic value used in $P = NpN_3Di^5$. It is not recalculated from blade geometry.

Initial agitation - 350 rpm

App assumption | Generic reusable glass profile

Rule: PROFILE_DEFAULTS["glass"].baseRpm = 350

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable glass starting profile. This is the initial agitation setting loaded into the form.

Maximum agitation - 1200 rpm

App assumption | Generic reusable glass profile

Rule: PROFILE_DEFAULTS["glass"].maxRpm = 1200

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable glass starting profile. This is the ceiling available to the generic DO-control cascade.

Initial aeration - 0.3 vvm

App assumption | Generic reusable glass profile

Rule: PROFILE_DEFAULTS["glass"].baseVvm = 0.3

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable glass starting profile. This is the initial gas rate loaded into the form.

Maximum aeration - 2 vvm

App assumption | Generic reusable glass profile

Rule: PROFILE_DEFAULTS["glass"].maxVvm = 2

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable glass starting profile. This is the gas-rate ceiling available to the generic DO-control cascade.

Baffled vessel - Yes

App assumption | Generic reusable glass profile

Rule: PROFILE_DEFAULTS["glass"].baffled = true

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable glass starting profile. This Boolean changes the generic vessel configuration and mixing interpretation.

Maximum pressure - 0.5 bar(g)

App assumption | Generic reusable glass profile

Rule: PROFILE_DEFAULTS["glass"].maxPressure = 0.5

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable glass starting profile. This is the pressure ceiling used for warnings in the generic equipment model.

Default operating pressure - 0 bar(g)

App assumption | Generic reusable glass profile

Rule: PROFILE_DEFAULTS["glass"].operatingPressure = 0

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable glass starting profile. This is the starting headspace pressure loaded into the form.

Maximum inlet oxygen fraction - 100%

App assumption | Fixed simulator capability assumption

Rule: maxOxygenFraction = 100

Selection / calculation: No calculation. Every vessel preset currently permits a 100% oxygen inlet ceiling in the generic gas cascade.

Limitations: This describes the simulator's permitted gas-mixture range, not proof that the selected hardware is certified for pure oxygen service.

Configuration notes and unresolved evidence

2026-09-11 field-level review: use the evidence beside each parameter; an archived document does not verify the whole preset.

The printed impeller/tank ratios do not all match the absolute dimensions. Direct millimetre dimensions are preferred.

No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Local original document identities

Sartorius Univessel Glass brochure SBI1554-e

sartorius-univessel-glass-brochure-sbi1554-e-fade553cc998.pdf

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